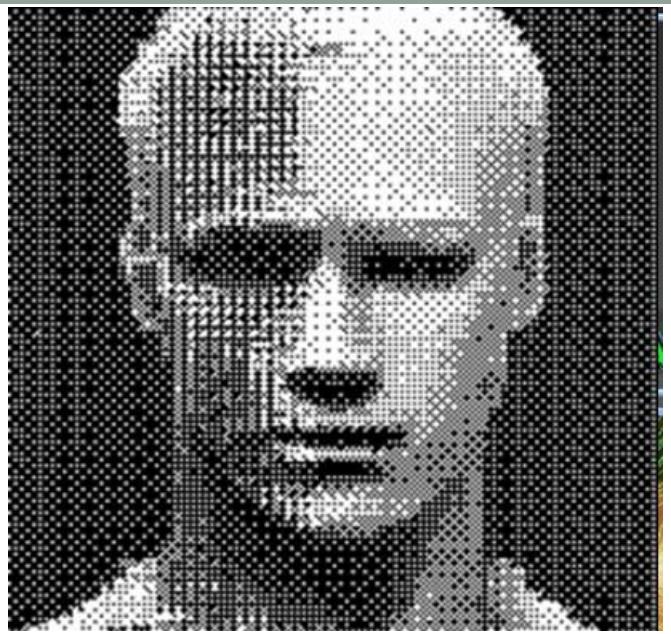


DIGITAL IMAGE PROCESSING

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CMPn332 - CMPN332 & CMP 432

What is the objective of image processing?

- **Image Analysis and Interpretation→** Extract meaningful information from an image for further analysis or decision-making by machines or humans.

- **Image Enhancement→** Geometric transformation**,** Increasing contrast, reducing noise, sharpening details, or adjusting brightness.

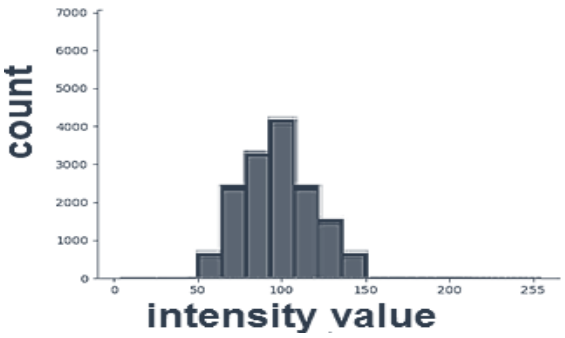
- **Image Restoration→** Restore image to its original or an improved state.

- **Image Compression→** Reduce the size of an image file for efficient storage or transmission without losing significant quality

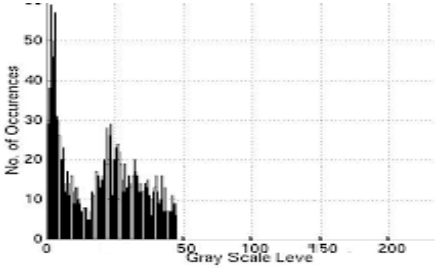
- **Image Segmentation→** Divide an image into meaningful regions or segments to isolate areas of interest for further analysis.

Histogram

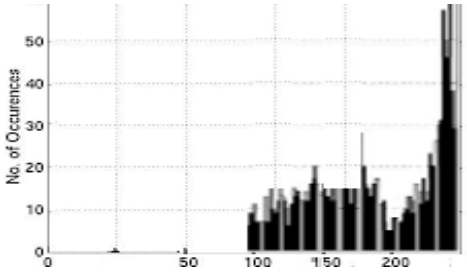
- Histogram is a graph, which gives you an overall idea about the intensity distribution of an image.



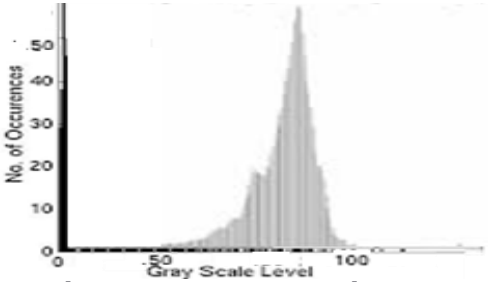
- It is a graph with pixel values (ranging from 0 to $2^{\text{pixel depth}}-1$) in X-axis and corresponding number of intensity count in the image on Y-axis.
- By looking at the histogram of an image, we get intuition about contrast, brightness, intensity distribution of that image.



Low brightness Image



High brightness Image



Low contrast Image

Histogram

Histogram Algorithm

Input: Image matrix

Start:

Initialize a frequency array of length based on the pixel depth with zero values

$$frequency[2^{pixel\ depth}] = \{0\}$$

for all pixels in the image do

frequency [pixel intensity value]+=1;

end for

Plot the Histogram:

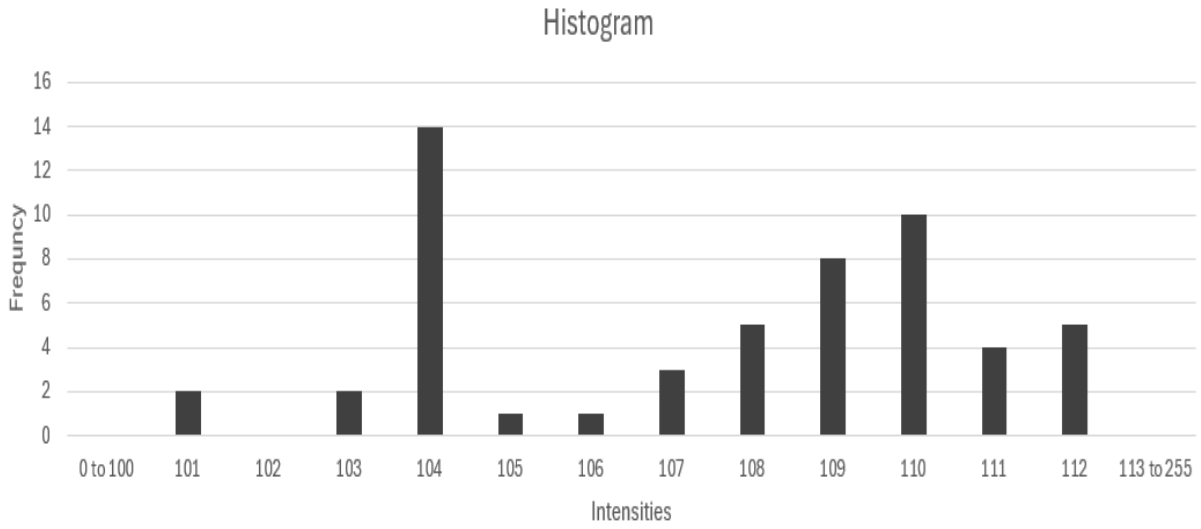
- Use the frequency counts and intensities range to plot the histogram
- Each intensity value is represented as a bar, the height of which corresponds to the frequency of data points in that intensity.

End

Histogram

- Example: A 8-bit 11 x 5 image shown in the following image matrix find the histogram?

```
Image Matrix =
[110, 103, 103, 104, 108;
 109, 101, 104, 104, 110;
 109, 109, 105, 104, 110;
 111, 111, 109, 104, 110;
 108, 109, 110, 112, 109;
 108, 109, 110, 108, 108;
 112, 110, 107, 110, 109;
 110, 107, 104, 104, 110;
 112, 112, 104, 104, 111;
 107, 101, 104, 104, 112;
 106, 104, 104, 104, 111]
```



Gray level	Frequency
0→100	0
101	2
102	0
103	2
104	14
105	1
106	1
107	3
108	5
109	8
110	10
111	4
112	5
113→255	0

What is the objective of image processing?

- **Image Analysis and Interpretation→** Extract meaningful information from an image for further analysis or decision-making by machines or humans.
- **Image Enhancement→** Geometric transformation**,** Increasing contrast, reducing noise, sharpening details, or adjusting brightness.
- **Image Restoration→** Restore image to its original or an improved state.
- **Image Compression→** Reduce the size of an image file for efficient storage or transmission without losing significant quality
- **Image Segmentation→** Divide an image into meaningful regions or segments to isolate areas of interest for further analysis.

Image Enhancement

- It modify pixel values without changing their spatial relationships.
- Image enhancement is performed in both the spatial domain and the frequency domain.
  - **Spatial Domain**: Works directly with pixel values, offering straightforward method to implement. Best for **local adjustments.**
  - **Frequency Domain**: Images are transformed using techniques like the Fourier Transform, which separates the image into its frequency components. Enhancement is then applied to these components, which represent different levels of detail in the image. Best for **global adjustments** and specific frequency manipulations.

Image Enhancement in spatial domain

Image enhancement in the spatial domain is categorized into two primary techniques:

Point processing

- Focuses on individual pixel manipulation through a transformation function that defines how each pixel's value will be modified based on its original value.

Neighborhood processing

- Focuses on modifying a pixel's value based on its neighbors. The pixel's new value is calculated based on the values of itself and its neighbors by using a kernel (mask).

Image enhancement Point processing

Key point processing techniques used in image enhancement are:

- **Histogram Equalization**
- **Contrast Stretching**
- Image Negative
- Logarithmic Transformation
- Gamma Correction(Power law transformation)
- Grey level slicing
- Bit plane slicing

Image enhancement Point processing

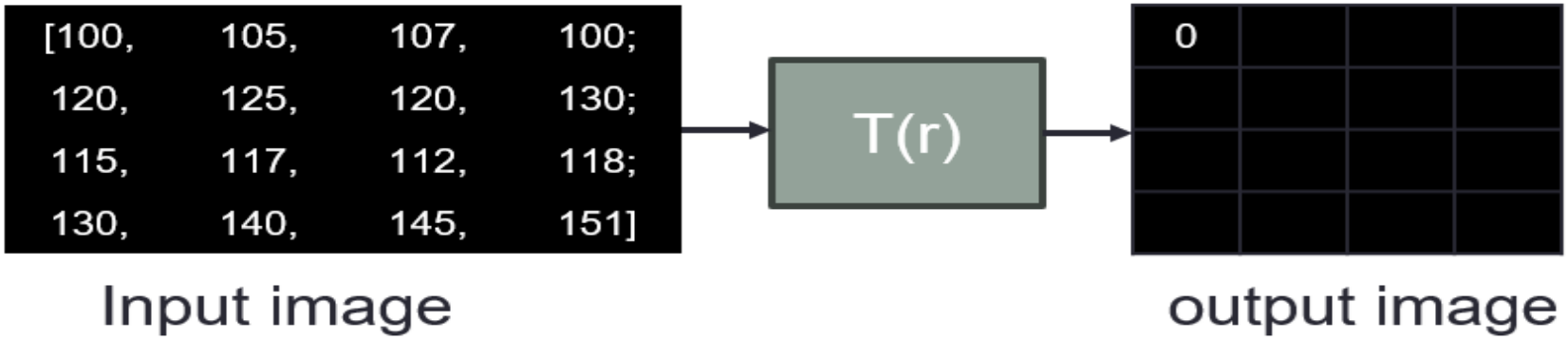
The simplest formula for image enhancement technique is:

$$S=T(r)$$

Where:

S: pixel value of output image after processing

T: transformation that maps image pixel r into pixel value of input image before processing **r:**

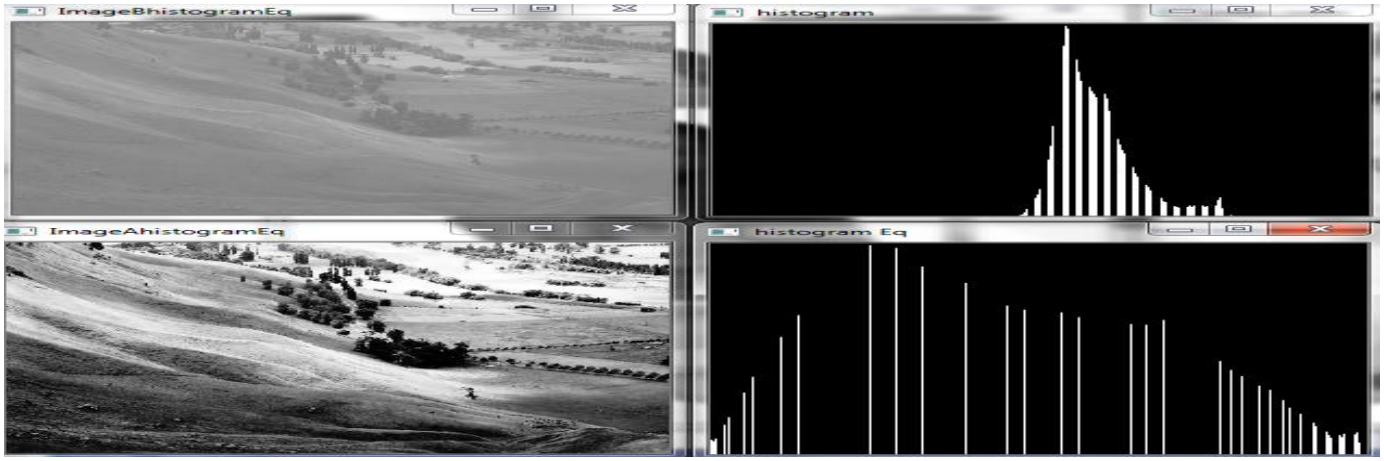


Histogram Equalization

- It Is a method which increases the dynamic range of the gray levels in a low contrast image to cover the full range of gray levels in order to improve image contrast.
- It is a simple way to improve dark or washed out images

$$S=T(r)$$

$$S=CDF(r)\times(L-1)$$



Histogram Equalization

- Example: 8-bit 5 x 11 image shown in the following image matrix. Find out the equalized histogram gray levels for the image.

```
Image Matrix =
[110, 103, 103, 104, 108;
 109, 101, 104, 104, 110;
 109, 109, 105, 104, 110;
 111, 111, 109, 104, 110;
 108, 109, 110, 112, 109;
 108, 109, 110, 108, 108;
 112, 110, 107, 110, 109;
 110, 107, 104, 104, 110;
 112, 112, 104, 104, 111;
 107, 101, 104, 104, 112;
 106, 104, 104, 104, 111]
```

Gray level	Frequency
101	2
102	0
103	2
104	14
105	1
106	1
107	3
108	5
109	8
110	10
111	4
112	5

Histogram Equalization

Gray level r_k	Frequency	Probability= $P_r(r_k)=\frac{count}{M*N}$	CDF=$\sum_{j=0}^{k} p_r(rj)$	Equalized grays= $(L-1)*CDF$
101	2	0.036364	0.036364	9
102	0	0	0.036364	9
103	2	0.036364	0.072727	19
104	14	0.254545	0.327273	83
105	1	0.018182	0.345455	88
106	1	0.018182	0.363636	93
107	3	0.054545	0.418182	104
108	5	0.090909	0.509091	130
109	8	0.145455	0.654545	167
110	10	0.181818	0.836364	213
111	4	0.072727	0.909091	232
112	5	0.090909	1	255

Histogram Equalization

```
Image Matrix =
[110, 103, 103, 104, 108;
 109, 101, 104, 104, 110;
 109, 109, 105, 104, 110;
 111, 111, 109, 104, 110;
 108, 109, 110, 112, 109;
 108, 109, 110, 108, 108;
 112, 110, 107, 110, 109;
 110, 107, 104, 104, 110;
 112, 112, 104, 104, 111;
 107, 101, 104, 104, 112;
 106, 104, 104, 104, 111]
```

```
Output Image Matrix=
 [213,   19,   19,   83,  130;
  167,    9,   83,   83,  213;
  167,  167,   88,   83,  213;
  232,  232,  167,   83,  213;
  130,  167,  213,  255,  167;
  130,  167,  213,  130,  130;
  255,  213,  104,  213,  167;
  213,  104,   83,   83,  213;
  255,  255,   83,   83,  232;
  104,    9,   83,   83,  255;
   93,   83,   83,   83,  232]
```

Histogram Equalization

A 5-bit 40x40 image has the following intensities, Find out the equalized histogram gray levels for the image.

Gray level	Frequency
11	580
15	320
10	150
20	250
18	300

- **Answer**:
- A 5-bit →$L=2^5 = 32$
- 40x40 → M*N=40*40=1600

Gray level r_k	Frequency	Probability= $P_r(r_k)=\frac{count}{M*N}$	CDF=$\sum_{j=0}^{k} p_r(rj)$	Equalized grays= $(L-1)*CDF$
10	150	0.093	0.093	3
11	580	0.362	0.456	14
15	320	0.2	0.656	20
18	300	0.187	0.843	26
20	250	0.156	1	31

The histogram equalization stretch out the intensity range from 10:20 to become 3:31 which improve the image contrast

Histogram Equalization

Histogram Equalization Algorithm

Input: Image matrix

Start:

Initialize a frequency array of length based on the pixel depth with zero values

frequency[$2^{\text{pixel depth}}$] = {0}

For each pixel in the image matrix do

frequency [pixel intensity value]+=1;

End for

Prob [$2^{\text{pixel depth}}$] = {0}

For i=0 to $2^{\text{pixel depth}}$ -1 do

Prob[i]= frequency [i]/ M x N

End for

CDF [$2^{\text{pixel depth}}$] = {0}

CDF [0] = Prob [0]

For i=1 to $2^{\text{pixel depth}}$ -1 do

CDF [i] = CDF [i-1] + Prob [i]

End for

//Remap normalized intensity to the image matrix

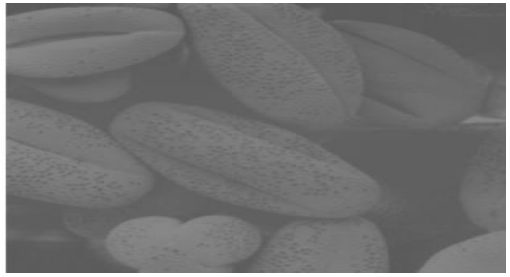
For each pixel in image matrix do

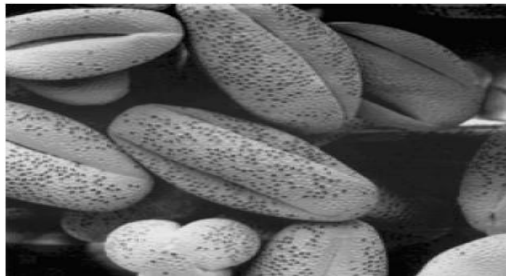
New_ pixel intensity value =round ($2^{\text{pixel depth}}$ -1*CDF[pixel intensity value]

End

Contrast stretching

- Low contrast image can result from:
  - Poor illumination
  - Lack of dynamic range
- The idea behind contrast stretching is to increase the dynamic range of the gray levels in the image
  - It means bright pixels in the image will become brighter and the dark pixels will become darker





Contrast stretching

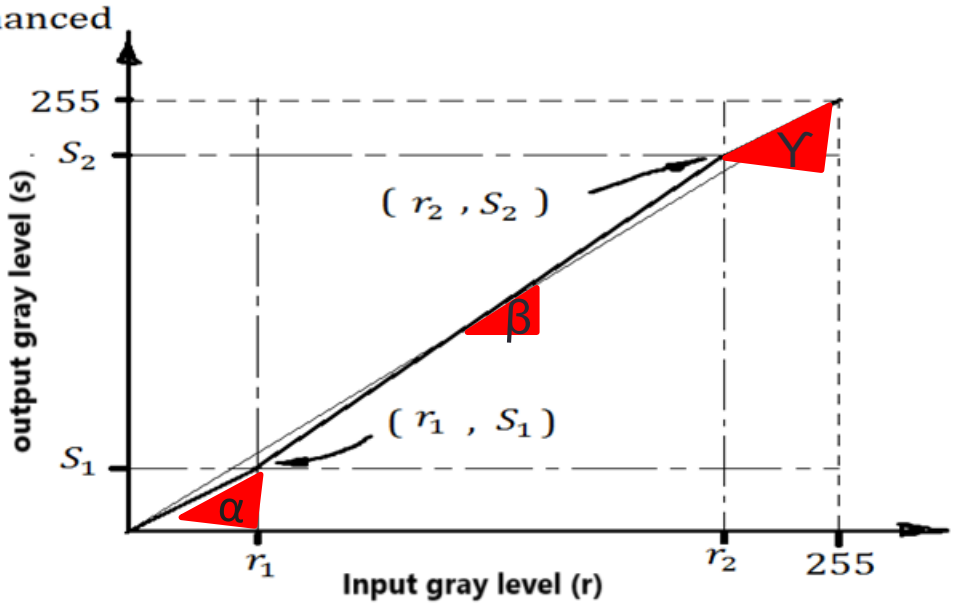
- The simple contrast stretching is increasing the Dynamic Range giving that the points (r_1, s_1) , (r_2, s_2) satisfying the following conditions:-

$$r_1 > s_1 \ , r_2 < s_2$$

Where r represent input values and s is the desired output

$$\alpha = \frac{s_1 - 0}{r_1 - 0}, \ \beta = \frac{s_2 - s_1}{r_2 - r_1}, \ \Upsilon = \frac{L - 1 - s_2}{L - 1 - r_2}$$

$$s = \begin{cases} \alpha * r & 0 \le \boldsymbol{r} < r_1 \\ \beta\,(r - r_1) + s_1 & r_1 \le \boldsymbol{r} < r_2 \\ \Upsilon(r - r_2) + s_2 & r_2 \le \boldsymbol{r} < L - 1 \end{cases}$$



Contrast stretching

- Example:
- A 3-bit 4x4 image has the following intensities, Find out the output gray levels after applying contrast stretching. Using r1=3, r2=5 s1=2 and s2=6

Answer:

[4,	3,	5,	2;
3,	6,	4,	6;
2,	2,	6,	5;
7,	6,	4,	1]

$$\alpha = \frac{s_1 - 0}{r_1 - 0} = 0.66,\ \beta = \frac{s_2 - s_1}{r_2 - r_1} = 2,\ \Upsilon = \frac{7 - s_2}{7 - r_2} = 0.5$$

$$s = \begin{cases} \alpha * r & 0 \le \boldsymbol{r} < r_1 \\ \beta\,(r - r_1) + s_1 & r_1 \le \boldsymbol{r} < r_2 \\ \Upsilon(r - r_2) + s_2 & r_2 \le \boldsymbol{r} < L - 1 \end{cases}$$

$$s = \begin{cases} 0.66 * r & 0 \le \boldsymbol{r} < 3 \\ 2 * (r - 3) + 2 & 3 \le \boldsymbol{r} < 5 \\ 0.5 * (r - 5) + 6 & 5 \le \boldsymbol{r} < 7 \end{cases}$$

[4,	2,	6,	1;
2,	7,	4,	7;
1,	1,	7,	6;
7,	7,	4,	1]

Contrast stretching

- Example: A 8-bit image with the following intensities, Find out the output gray levels after applying contrast stretching. Using the image dynamic range to represent r1,r2 and full dynamic range to represent s1 and s2

Answer:

r1=100, r2=151, s1=0, s2=255

$\alpha = \frac{s_1-0}{r_1-0} = 0$, $\beta = \frac{s_2-s_1}{r_2-r_1} = 5$, $\Upsilon = \frac{255-s_2}{255-r_2}$=0

$$s = \begin{cases} \alpha * r & 0 \le \boldsymbol{r} < r_1 \\ \beta\,(r - r_1) + s_1 & r_1 \le \boldsymbol{r} < r_2 \\ \Upsilon(r - r_2) + s_2 & r_2 \le \boldsymbol{r} < L-1 \end{cases}$$

$$s = \begin{cases} 0 & 0 \le \boldsymbol{r} < 100 \\ 5 * (r - 100) & 100 \le \boldsymbol{r} < 151 \\ 255 & 151 \le \boldsymbol{r} < 255 \end{cases}$$

[100,	105,	107,	100;
120,	125,	120,	130;
115,	117,	112,	118;
130,	140,	145,	151]

[0,	25,	35,	0;
100,	125,	100,	150;
75,	85,	60,	90;
150,	200,	225,	255]

Histogram equalization vs Contrast stretching

- Although both histogram equalization and contrast stretching enhance image contrast, they do so in fundamentally different ways:
- **Histogram Equalization**:
  - It's non-linear transformation. Redistributes pixel intensities across the entire range of available values based on the cumulative distribution function (CDF). This can lead to a more uniform histogram and enhanced visibility of features across the image but Once the original pixel values are modified, the specific mapping used cannot be reversed to retrieve the original pixel values because it involves a non-linear transformation.
- **Contrast Stretching**:
  - It's linear transformation. Involves linearly mapping pixel values from a specified range to a wider range.
  - This method adjusts the contrast while preserving the overall distribution of the original pixel values, focusing on enhancing specific regions without affecting others too.
- **Histogram equalization is better for enhancing features in low-contrast images, while contrast stretching is preferred for images with known intensity ranges where a natural look is important.**

Contrast stretching_ Solve

In which scenario would you typically use histogram equalization instead of contrast stretching?

A) When the image has a very limited dynamic range

B) When you want to enhance specific features of the image

C) When the histogram is already uniformly distributed

D) When you need to maintain the original brightness

Answer: A) When the image has a very limited dynamic range


Thank you!